

Lemma. Let $X: \mathcal{U} \subset \mathbb{R}^2 \rightarrow S$ be a parametrization of a regular surface S .
Given $q \in \mathcal{U}$,

$$DX_q[\mathbb{R}^2] = \{ \alpha'(0) \mid \alpha: (-\varepsilon, \varepsilon) \rightarrow S \text{ is differentiable and } \alpha(0) = X(q) \}.$$

Proof. Let w be a tangent vector at $X(q)$. That is,
for some smooth curve $\alpha: (-\varepsilon, \varepsilon) \rightarrow X[\mathcal{U}] \subset S$ such that $\alpha(0) = X(q)$, $w = \alpha'(0)$.
Since $X^{-1}: X[\mathcal{U}] \rightarrow \mathbb{R}^2$ is differentiable, the curve $\beta = X^{-1} \circ \alpha: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}$ is also
differentiable, so $DX_q \cdot \beta'(0) = DX_q \cdot DX_{X(q)}^{-1} \cdot \alpha'(0) = \alpha'(0)$.

Conversely, let $w = DX_q(V)$ for some $V \in \mathbb{R}^2$. Then, a curve

$$\gamma: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}: t \mapsto tV + q$$

satisfies that $\gamma'(0) = V$ and $\gamma(0) = q$. Now, a curve $\alpha = X \circ \gamma$ satisfies
 $\alpha(0) = X(q)$ and $\alpha'(0) = DX_q \cdot V = w$, so proved. $\square$